AL SAFA 2 PARK

Dubai Municipality — AI-Powered Park Design Challenge

DELIVERABLE 1

Site Analysis & Human-Centric Research

Prepared by: Assim
4th-Year Student, AI & Data Science — aivancity
July 2026

Corresponds to Scope of Work Sections: "Site Understanding and Context Analysis" and "User and Community Analysis"

How to Read This Document

This report distinguishes between three levels of certainty throughout, so the design decisions it supports are traceable back to their evidence — a deliberate methodological choice given this challenge's emphasis on AI-driven, data-grounded design reasoning.

[CONFIRMED] Verified against the official as-built CAD file, official brief documents, or independently cross-checked real-world data (OpenStreetMap, live location data, geodetic calculation).

[INFERRED] A reasoned conclusion built from confirmed data, but not independently verified — presented as a working hypothesis for design purposes.

[GAP] Explicitly missing information. Flagged rather than silently filled with assumption.

Full source-by-source methodology and AI tool usage log for this analysis is documented in Deliverable 10 — AI Methodology Report.

Section 1 — Site Context & Opportunities Assessment

1.1 Site Identification

Attribute	Value
Site name	Al Safa 2 Park
Address	11 31A Street, Al Safa Second, Al Safa, Dubai, UAE
Coordinates	25.155833°N, 55.221764°E
Site area (brief-stated)	15,000 m²
Site area (measured from confirmed boundary)	14,841 m² (within 1% of brief figure)
Boundary verification method	As-built DWG geometry, cross-matched to OpenStreetMap tagged park polygon (way 87408738) and to embedded DWG coordinate text via independent UTM Zone 40N conversion

1.2 Site Geometry & Physical Characteristics

The site is a regular rectangle — confirmed geometrically (interior angles 89.3°–91.1°, i.e. 90° within survey/drafting tolerance), not merely close to one.

Attribute	Value
Shape	Rectangle, 4 vertices, 90° corners
Long edges	166.0 m (NE) / 163.8 m (SW)
Short edges	87.5 m (NW) / 91.8 m (SE)
Total area	14,841 m²
Existing trees / shrubs on site	75 (generic block symbol only — species and canopy size not recoverable from source file)
Topography	Not present in source file — site is within Dubai's generally flat inland terrain;

Attribute	Value
	treated as effectively level pending on-site confirmation

1.3 Urban Context & Adjacency

Confirmed directly from the official as-built drawing's text annotations:

Adjacent Use	Relevance to Design
Shopping Center	Short-visit / respite users; lunch-break traffic
Aisha Butti Al Mulla Masjid (mosque)	Pre/post-prayer social use; acoustic sensitivity on that edge
Umm Suqeim Model School (intermediate)	School-age (approx. 10–14) after-school user surge, weekday afternoons
Dubai Center for Physiotherapy and Rehabilitation	Daily mobility-impaired / elderly users — a live accessibility user group, not a hypothetical one
Surrounding villa plots	Resident families — existing, repeat, multigenerational user base
Road frontage	18.29 m and 36.58 m road classes, plus a 6.5 m pedestrian "sikka" — multiple access typologies available
Public parking (~4 areas)	Distributed vehicular access in surrounding blocks

1.4 Existing Conditions & Assets

The site is a currently operating public park, not a blank or derelict site — confirmed via live public location data (ratings and reviews), independent of the official brief documents. Existing informal amenities include a 5-a-side football ground, a basketball court, a sand volleyball court, and drinking-water facilities. Multiple reviewers describe repeat family use and value the park as a genuine neighborhood asset.

[GAP / OPPORTUNITY] One recent review reports courts being restricted to residents of the immediate neighborhood only, describing this as a new and frustrating policy. This is a real, current equity issue in the existing park's operation. The redesign should address this directly — e.g. a clearly stated access policy, or space that is inherently harder to gatekeep than single-purpose courts — rather than silently preserving the status quo.

[CONFIRMED] A small existing food-and-beverage kiosk, "Eat and Relax Cafeteria," sits within the park boundary itself (confirmed via geolocation, tagged shop=convenience). This is a real existing commercial asset to evaluate for retention/upgrade under the brief's required Commercial and Service Facilities component, rather than a function to introduce from scratch.

1.5 Orientation & Environmental Analysis

True north was not present in the source CAD file. It was established independently by querying OpenStreetMap for the tagged park polygon, matching its four edges to the CAD boundary by length (agreement within 1.5° across all four edges — within OSM's own community-tracing tolerance), and computing the required rotation. Result: the CAD file's coordinate axes are already aligned to true north/east to within approximately 0.5° — no meaningful rotation correction is required.

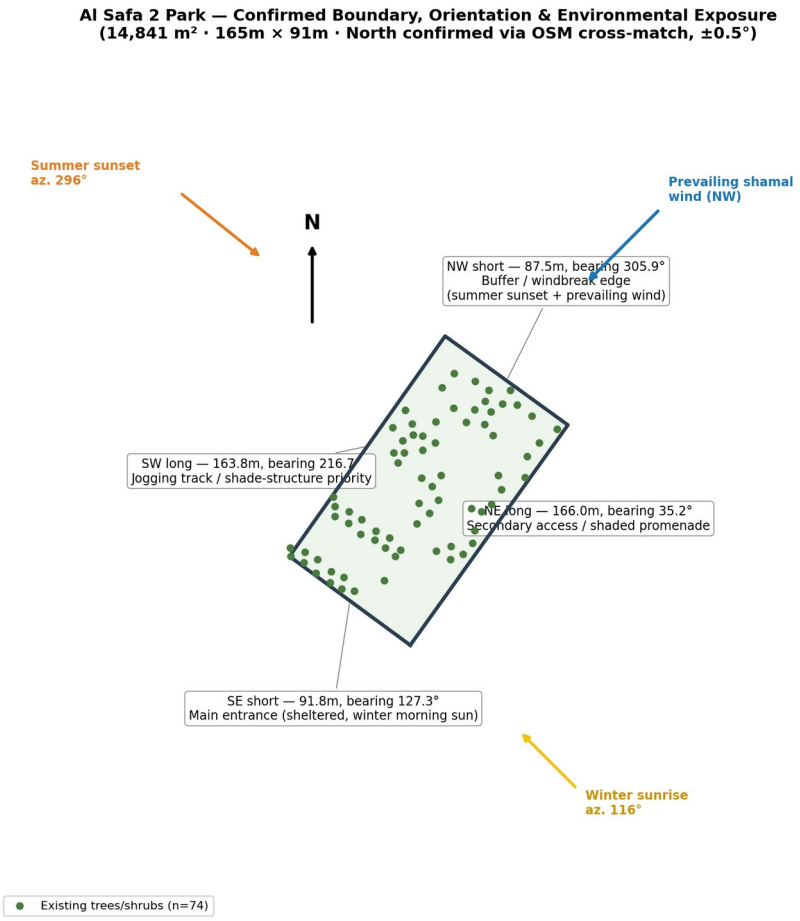


Figure 1.1 — Confirmed boundary, true-north orientation, edge bearings, existing trees, and combined solar/wind exposure read per edge.

Edge	Bearing	Length	Confirmed Exposure	Design Read [INFERRED]
NW (short)	305.9°	87.5 m	Summer sunset (az. 296°) + prevailing shamal wind (NW)	Buffer / windbreak planting; service functions. Least suitable for primary gathering.
SE (short)	127.3°	91.8 m	Winter sunrise (az. 116°); sheltered from prevailing wind	Best candidate for main entrance / arrival plaza; morning activation.
NE (long)	35.2°	166.0 m	Morning sun; partial wind shelter	Secondary access; shaded promenade.
SW (long)	216.7°	163.8 m	Winter afternoon sun (az. 213–235° at 2–4pm); summer transitional exposure	Jogging track / shade-structure density priority.

Regional climate parameters (Dubai, hot desert / Köppen BWh), sourced independently to ground the sustainability and shading strategy:

Parameter	Value
Summer daytime temperature	40–41°C average highs, occasionally exceeding 49°C
Winter daytime temperature	~24°C average
Relative humidity	~61% (winter peak) to ~42% (spring low); summer dew points 24–28°C

Parameter	Value
	(persistently muggy)
Prevailing wind	North-west ("shamal"); can arrive from any direction during storm events
Sunshine	Approx. 9–10.5 hours/day year-round

1.6 Opportunities & Constraints Matrix

Opportunities	Constraints
Existing informal sports use signals real, provable demand — a base to formalize rather than invent	Reported resident-only court access — an equity issue to resolve, not replicate
Adjacent school + rehab center = a guaranteed daily user base across two distinct, high-need groups	Adjacency to the mosque requires acoustic/activity buffering on that edge
Regular rectangular geometry — efficient to zone and program	Elongated 166×88 m proportions constrain how zones can be arranged — long, narrow logic, not a flexible blob
NW wind could support passive cooling in the right locations	Same NW wind carries shamal dust — needs filtering/buffering, not open exposure
Confirmed true-north orientation enables a genuinely edge-specific (not generic) shade and zoning strategy	No topographic or subsurface infrastructure data exists in the source file — will need on-site or utility-provider confirmation before final engineering

Section 2 — User & Community Analysis

2.1 Key User Groups & Stakeholders

Derived directly from confirmed site adjacencies — each neighboring use generates a specific, predictable user group rather than a generic "public":

User Group	Source of Confirmation	Primary Need
School-age youth (approx. 10–14)	Adjacent Umm Suqeim Model School	After-school sport, informal social space, weekday afternoon peak
Mobility-impaired / elderly patients	Adjacent Dubai Center for Physiotherapy and Rehabilitation	Level, shaded, rest-frequent circulation suited to therapeutic walking
Worshippers	Adjacent Aisha Butti Al Mulla Masjid	Quiet buffer zone; brief post-prayer gathering space
Shoppers / workers	Adjacent Shopping Center	Short-visit respite; shade and seating over destination programming
Resident families	Surrounding villa plots + live review evidence of repeat use	Existing daily/weekly users; multigenerational, safety-conscious
Informal sport players	Live review evidence — existing football / basketball / volleyball use	Court quality, booking fairness, flexible active space

2.2 User Personas [INFERRED — see note]

[INFERRED] The five personas below are structured from confirmed adjacency data and real review evidence, not from direct interviews or surveys. They are a working starting point, not validated research — see Section 2.6.

Amina, 12 — Student

Walks over from the adjacent school with friends 2–3x/week after class. Wants informal sport and hangout space; has an existing sense of ownership over the park (echoed in the resident-access complaint below).

Rashid, 68 — Physiotherapy outpatient

Visits the adjacent rehab center twice weekly, uses a walking frame. Currently has no reason to enter the park after appointments. A shaded, level loop path could convert an adjacent captive population into active daily users — a real, unclaimed opportunity, not a checkbox.

Fatima, 34 — Parent, nearby villa

Two children, ages 4 and 8. Weekday-morning and weekend visitor. Needs clear sightlines between play areas and seating, shade, and nearby drinking water.

Yusuf, 24 — Informal athlete

Plays football/basketball 3x/week. Prioritizes court quality and fair, transparent access over aesthetics.

Mariam, 45 — Worshipper

Walks to Friday prayers at the adjacent mosque. Values a quiet buffer near that edge and a calm space to sit briefly afterward.

2.3 Community Aspirations & Real-World Evidence

[CONFIRMED] The park is already a valued, actively used neighborhood asset — repeat family visits and organized informal sport are documented in live public reviews, independent of any brief document.

[INFERRED] The redesign's role is to upgrade, rebalance, and open up an already-loved space (shade, equity, inclusivity, formalized programming) rather than to manufacture demand from zero. This reframes the design narrative away from generic "activation" language toward a more specific, evidence-backed story.

2.4 Accessibility & Inclusivity Strategy

The brief requires accommodation of children, youth, families, older adults, and People of Determination, in line with "national and international accessibility standards." The applicable emirate-level standard is the Dubai Universal Design Code — issued by Dubai Municipality itself, the body running this challenge, and covering public-space urban design elements including pedestrian routes, playgrounds, and wayfinding, underpinning the emirate's "Wosool" accessibility certification.

Given the confirmed, daily-use adjacency to a physiotherapy and rehabilitation center, accessibility here is a functional design requirement grounded in a real, present user group — not a generic compliance checkbox. Working priorities:

- Level, non-slip, adequately wide primary circulation paths
- Frequent shaded rest points on any loop intended for therapeutic or leisure walking
- Tactile wayfinding at key decision points
- An uninterrupted accessible route from the sheltered SE entrance through to core amenities — no gradient or surface discontinuity

2.5 Human-Centered Design Principles for This Site

Synthesized from the brief's stated design requirements and applied to the confirmed site-specific findings above, rather than restated generically:

- Intergenerational by adjacency, not just by intention — the site's neighbors already guarantee a cross-generational user mix; design should physically connect rather than segregate their zones.
- Equity as an explicit design response to a documented, current problem (court-access gatekeeping), not an abstract value statement.
- Edge-specific comfort strategy — each of the four boundary edges has a distinct, confirmed sun/wind exposure profile; shade and programming intensity should follow that, not be applied uniformly.
- Accessibility as a daily-use design driver, given the confirmed adjacent rehabilitation center, not a retrofit consideration.

2.6 Research Validation Gap & Recommendation

[GAP] No direct primary research (interviews, surveys, or on-site observation) has been conducted. Every conclusion in Section 2 is built from confirmed adjacency data, published standards, or real but indirect evidence (public reviews) — not from asking residents, students, or patients directly.

If time permits before submission, even a lightweight validation pass would materially strengthen this deliverable: 30–45 minutes on-site with 8–10 informal conversations across the user groups identified in 2.1, or a short intercept survey near the existing courts. This is flagged as a genuine open recommendation, not a step being skipped silently.

[AUTHOR CONTEXT] The author has personal familiarity with this neighborhood — having studied in the area and maintaining friendships with residents of Al Safa Second. This is disclosed as relevant context: informal, lived familiarity with the community, not a structured research method, and not a substitute for the primary validation recommended above.

Sources: Dubai Municipality AI Park Design Challenge — Scope of Work, General Terms & Conditions; Neighborhood Parks Manual; official as-built AutoCAD drawing; OpenStreetMap (Overpass API); live public location data; Dubai Universal Design Code; independent climate data review. Full AI-tool-by-tool methodology log: Deliverable 10 — AI Methodology Report.